From Safety to Retraction: What ARA's Unclosed T2 Test Does—and Does Not—Show

1 November 2026

Abstract

Does failure of worldly-information safety require a speaker to retract an earlier epistemic-modal assertion? A stronger proposal, T2, makes such failure both necessary and sufficient for obligatory retraction while holding the original use context fixed. ARA tested this proposal and its two directions against a frozen representation of the literature. Its focused test returned six OPEN routes and no accepted inference. This paper explains that outcome without treating unsuccessful search as a refutation. A necessary condition on knowledge does not itself connect an unsafe belief to a duty concerning a particular assertion. A one-way retraction rule does not establish that its trigger exhausts all relevant obligations. The literature already supplies substantial parts of the surrounding debate; a bounded dossier classified only the exact biconditional as not found within its search scope. The resulting task is to supply and defend the missing connections, with belief, assertion, truth, assessment and obligation kept distinct. Later construction retained the relevant safety and retraction anchors, but the packed continuation completed 46 jobs with all routes OPEN. The accepted consolidation subset likewise retained no symbols shared across source papers. The missing connection remains a philosophical problem; a separate T2 proposal paper now argues for it explicitly.

1. The question and the status of this paper

An earlier assertion that something might be the case can become difficult to stand behind after further information arrives. Several explanations compete: the assertion may be false from a later assessment context; the speaker may lose knowledge; the conversational update may no longer be acceptable; or later responses may help determine what was originally said. These explanations do not identify the same event or impose the same norm. L3 is the project's label for a proposed connection between worldly-information safety in an account of modal knowledge and an assessment-sensitive obligation to retract. It is a research question, not a result established by naming both literatures. [1, 2, 14]

The candidate T2 strengthens that connection. For a particular earlier might-assertion, with its original use context held fixed, a later assessment requires retraction if and only if the associated original belief fails worldly-information safety as evaluated there. Thus unsafe belief would suffice for obligation, and safety preservation would exclude obligation. The thesis comes from the literature analyst's dossier. It was supplied to ARA as a disclosed external hypothesis, not discovered by its search and not added as a premise. [14]

This analysis locates the inferential work that the recorded test left undone and identifies what would address it. It does not supply a proof of T2 or establish literature priority.

2. Four distinctions the proposed bridge must preserve

First, a belief that might p, an assertion of might p, and a claim to know that might p are different objects. Beddor and Goldstein's information-sensitive safety condition constrains when a belief amounts to knowledge. In their formulation, nearby worlds at which the belief is held must make it true relative to the worldly information there. Their later presentation separately states belief, truth and safety conditions for probabilistic knowledge. A claim about loss of knowledge therefore cannot be silently transferred to the truth of the embedded modal assertion. [1, PDF pp. 5, 23–24]

Second, the context of the earlier assertion must remain identifiable. MacFarlane's Retraction Rule concerns an unretracted assertion made in one context and evaluated from another. The rule requires retraction when its content is not true as used at the original context and assessed from the later one. Changing the content, the original use context, or the object being withdrawn changes the proposal under examination. [2, printed pp. 108–109; PDF pp. 122–123]

Third, permission to reject, an obligation to retract, an actual retraction, and withdrawal of a knowledge attribution must be separated. MacFarlane explicitly distinguishes a sufficient condition for permissible rejection from a sufficient condition for obligatory retraction. Even regarding retraction as a special case of rejection does not turn permission into obligation. Observing an audience's rejection, or explaining a speaker's willingness to withdraw, accordingly leaves a further normative question. [2, printed p. 111; PDF p. 125]

Fourth, worldly information and a later assessor's information cannot be identified without an argument. Beddor and Goldstein discuss an interpretation using provisionally revised conversational or assessor credences, but the dossier correctly distinguishes that option from replacing a use-context parameter with a later assessment parameter. A proposed T2 proof must say whose belief is being tested, which information determines safety, and why that evaluation bears on this speaker's duty about this earlier assertion. Otherwise two formally similar expressions may concern different subjects. [1, PDF pp. 28–29; 14, T5]

These distinctions constrain criticism as well as defense. MacFarlane's allowance that one may retract something still thought true does not, by itself, refute a thesis about obligatory retraction. Nor may Hawke's criticisms of a safety variant without a truth conjunct be transferred without qualification to Beddor and Goldstein's original formulation. Hawke explicitly announces that alteration in the appendix he tests. [2, printed p. 109; 4, PDF p. 29]

3. What the T1–T9 literature dossier establishes

The dossier searched a bounded window from 2018 through 29 October 2026, using MacFarlane's 2014 book as an explicit earlier anchor. Its nine candidate formulations are analyst-generated possibilities for comparison with later ARA output. Four are classified as anticipated, four as partially anticipated, and one as not found. These labels concern the formulations and the inspected evidence; they do not certify truth, viability, consensus or exhaustive priority. Seven further lawful full texts were acquired during that dossier round. Acquisition did not insert their claims into ARA's existing freeze. [14]

ID	Proposed connection or rival	Dossier verdict	What remains to be distinguished
T1	Safety failure suffices for obligatory retraction	Partially anticipated	The ingredients exist; the normative bridge was not established.
T2	Safety failure is the exact criterion for obligatory retraction	Not found within scope	Both directions and their common belief/assertion/context parameters need support.
T3	Asymmetric loss explains withdrawal without relativism	Anticipated	Behavioral explanation of knowledge-attribution judgments is not a bare-assertion duty.
T4	Stable acceptance guides withdrawal	Partially anticipated	Denying knowledge now is not yet a fixed-use retrospective obligation.
T5	Make worldly information assessment-relative	Partially anticipated	An assessor-credence interpretation does not itself license parameter substitution or retraction.
T6	Flexible truth assessment with current-context retraction	Anticipated	Truth judgments and acceptance/retraction judgments can have different context dependence.
T7	Retraction cancels, or proposes to cancel, speech-act effects	Anticipated	Cancellation, uptake and obligation must not be conflated.
T8	Later responses help determine original content	Anticipated	This metasemantic account changes the fixed-content explanatory target.
T9	Fragmented, topic-sensitive knowledge replaces safety	Partially anticipated	The semantic rival exists; the added retraction obligation remains unverified.

The rivals matter because a superficial bridge can repeat an existing explanation. Dinges develops an asymmetric-loss account of knowledge-attribution judgments and explicitly extends the relevant epistemic qualities to safety and reliability. That is precedent for an explanatory connection involving safety; it does not already establish the requested obligation to retract a bare might-assertion. [5, PDF pp. 10–11]

Hawke's stable-acceptance account provides conditions for denying knowledge when available evidence fails to support the content. It also leaves questions about the contextual treatment of availability open. Fragmented and topic-sensitive alternatives supply a different account of modal knowledge. Neither simply mentioning an earlier assertion nor adding a retraction predicate establishes a justified extension of those theories. [3, PDF pp. 19, 22; 4, PDF pp. 29–32; 11; 14, T4, T9]

Other proposals relocate the explanation. Beddor and Egan distinguish flexible truth-value assessment from current-context acceptance and retraction. Rudin distinguishes rejection from judgments of falsity. Caponetto analyzes retraction through the normative effects of speech acts, while Bussière-Caraes and colleagues distinguish a proposal to undo effects from its successful uptake. Ball's temporal externalism allows later conduct to help determine earlier content. These positions make the question more precise: what does worldly-information safety add that the already available pragmatic or metasemantic account does not? [6, PDF pp. 14–18; 7, PDF pp. 7–9; 8, §§2–3; 9, §§7.3–7.4; 10, §§2–3]

The dossier's not-found verdict for T2 must be applied narrowly. Its necessity-only probe is a partial match to the biconditional. Its sufficiency-only probe is closest to T1 and is partially anticipated. A negation of a T2-related target does not inherit literature support merely because the positive thesis was not located. Further, accepting a safety condition need not entail contextualism, as Ask Zaar argues; Holliday and

Mandelkern explicitly identify truth, correctness and retraction as further questions for their framework. These are reasons to state each connection, not to treat the entire surrounding subject as untouched. [12, PDF pp. 1, 10–11; 13, PDF p. 58; 14; 14]

4. What ARA actually tested

ARA separates a bank of permitted inference forms, a frozen set of source-linked commitments, and proposed targets. A target's admission is not a proof: its subsequent search must supply licensed steps and satisfy the applicable critical questions. The experiments considered here reused the authenticated gap1r freeze: 160 commitments, comprising 159 source commitments from 22 of 26 input papers and one model-alignment bridge. The bridge was separately labelled and substantively disputed. The source coverage threshold was met, but the literal all-commitments stance criterion was not: only 159 of 160 entries had one of the three stipulated source stances. [14]

The old construction accepted 53 windows and skipped 21: 16 closure-bound failures, four formalization failures and one no-topical-roots outcome. This matters because the most relevant rule could be available in the rich extraction but absent from the smaller frozen T. A literature collection, an extracted graph and an admitted premise set are three different evidence inventories. Later acquisition or reading did not retroactively repair that freeze. [14]

Fork A supplied a two-hop target context with three full and 157 compact L3 commitments; every frozen formula and stance remained represented. Fork B supplied all 160 full commitments and provenance with a larger run-level context setting. Fork C then supplied the exact external T2 biconditional and its separate directions in a subsequent focused test. The arms therefore did not form a controlled test differing only in provenance length. [14]

Experiment	Targets	OPEN routes	CLOSED-derived	Terminal inference evidence
A: packed context	27	54	0	All terminal graphs are single open roots.
B: full context	20	40	0	One three-node partial sign argument; 39 single open roots.
C: external T2 and directions	3	6	0	All six terminal graphs are single open roots.

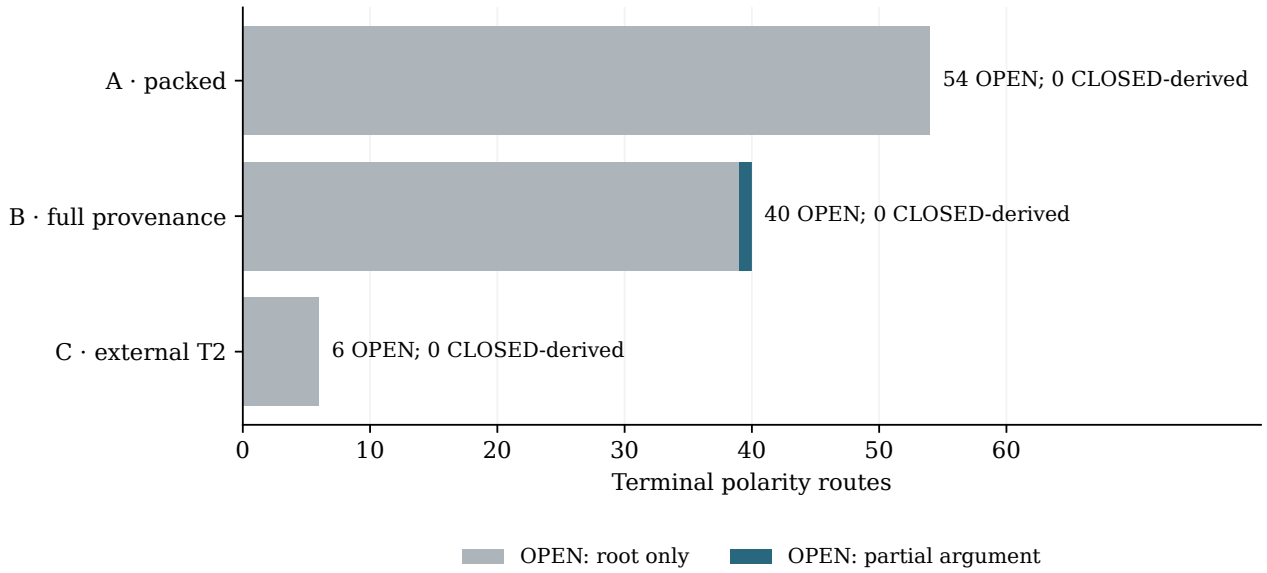


Figure 1. Recorded A/B/C terminal outcomes. B alone retains one partial sign argument, which is not L3 and remains OPEN. C tests the externally supplied biconditional and each direction; its six routes contain no accepted inference. Counts do not measure novelty or establish a causal effect of context length.

The partial B argument concerns a comparative assessment of relativism, not L3. Its graph retains a sign inference and two source leaves. One critical question had support; another lacked a premise for the combined comparative assessment, so the route stayed OPEN. One source conclusion also carried an omitted supporting inference. That route is a genuine partial-argument artifact, but neither closure nor the missing underlying source argument can be supplied by prose presentation. This comparative argument is distinct from the T2 proposal discussed below. [14]

A and B differed in ten semantic classifications before target packing despite identical pair inputs, and their configured bounds also differed. A retained more targets, but neither produced a demonstrated contribution. One sample per arm and the earlier discovery variation prevent attributing a philosophical quality effect to context size alone. [14]

C is the most direct test of T2, but has a crucial representation limit. The exact statements were represented by fresh whole-statement atoms; the biconditional and contextual quantification were not decomposed. Its directional atoms were not linked by a formal equivalence axiom. All six polarity searches stopped at open roots, with no accepted cut, support commitment, inference or critical-question answer. The experiment therefore records bounded failure under that representation, not a countermodel, a full logical analysis of T2, or an impossibility theorem about every licensed derivation. [14]

5. Why sufficiency and necessity need different premises

5.1 Sufficiency: from unsafe belief to a duty about an assertion

A compact reconstruction exposes the first missing step. Let K mean that the relevant belief is knowledge, S that it satisfies the relevant safety condition, F that the earlier assertion is not true as used originally and assessed later, and R that the speaker is required to retract that assertion. These are explanatory abbreviations, not ARA's actual formalization or a replacement semantics for the cited authors.

The safety constraint gives a necessary condition, K → S. Even accepting its classical contrapositive, ¬S → ¬K, yields loss of knowledge rather than F or R. MacFarlane's retraction rule supplies F → R for the appropriately specified assertion and contexts. Combining the two rules leaves the path from ¬S to F, or directly from ¬S to R, unfilled. A source-supported argument must supply that path; co-citing the two rules does not do so. [1, PDF p. 5; 2, printed p. 108]

A proponent could attempt a truth-mediated bridge: prove that the relevant safety failure makes the very same earlier assertion untrue from the specified assessment context, then apply the Retraction Rule. Alternatively, the proponent could defend an additional epistemic norm that requires retraction without that falsity claim. These are alternatives for investigation. Neither is a premise established by this paper, and the second would require explaining how the proposed duty relates to the assessment-sensitive theory rather than quietly replacing its trigger.

In the old frozen T, the complete general Retraction Rule was missing. The focused targets instead nominated related passages, including a supervaluationism example that did not supply a general safety-to-retraction rule. Nominating a passage as a basis did not establish its use in a licensed proof. The general rule was available in the rich extraction at physical PDF page 122 and was later retained in gap3 (§6). Its inclusion supplies an essential source premise without completing the normative bridge. [2; 14]

5.2 Necessity: the exhaustiveness problem

The converse demands a different kind of justification. T2 requires R → ¬S, or, classically, that safety preservation exclude the relevant obligation. F → R does not license its reversal. Even establishing ¬S → R would provide only sufficiency. A proof of necessity must either derive every relevant obligation's dependence on safety failure directly or show that an exhaustive account of its grounds entails that dependence.

The word "exhaustive" has substantive work to do. A norm that supplies one sufficient trigger leaves open whether other grounds generate the same obligation. Conversely, showing that a particular trigger fails does not show that no duty remains. T2's defender must specify the class of obligations being quantified over—constitutive assertoric duties, moral duties, or another restricted class—and justify the exclusion of alternatives within that class. Narrowing the class may yield a more defensible thesis, but it must be disclosed as a change of scope rather than credited as the original biconditional.

Two schematic valuations illustrate only the logical incompleteness of the displayed rules. With K false, S false, F false and R false, both K → S and F → R hold, while ¬S → R fails. With K false, S true, F false and R true, both rules again hold, while R → ¬S fails. These are truth-table witnesses about that small propositional skeleton, not models of the full philosophical theories, realistic cases, or ARA-produced countermodels. Their limited point is that the omitted links cannot be obtained from the two displayed implications alone.

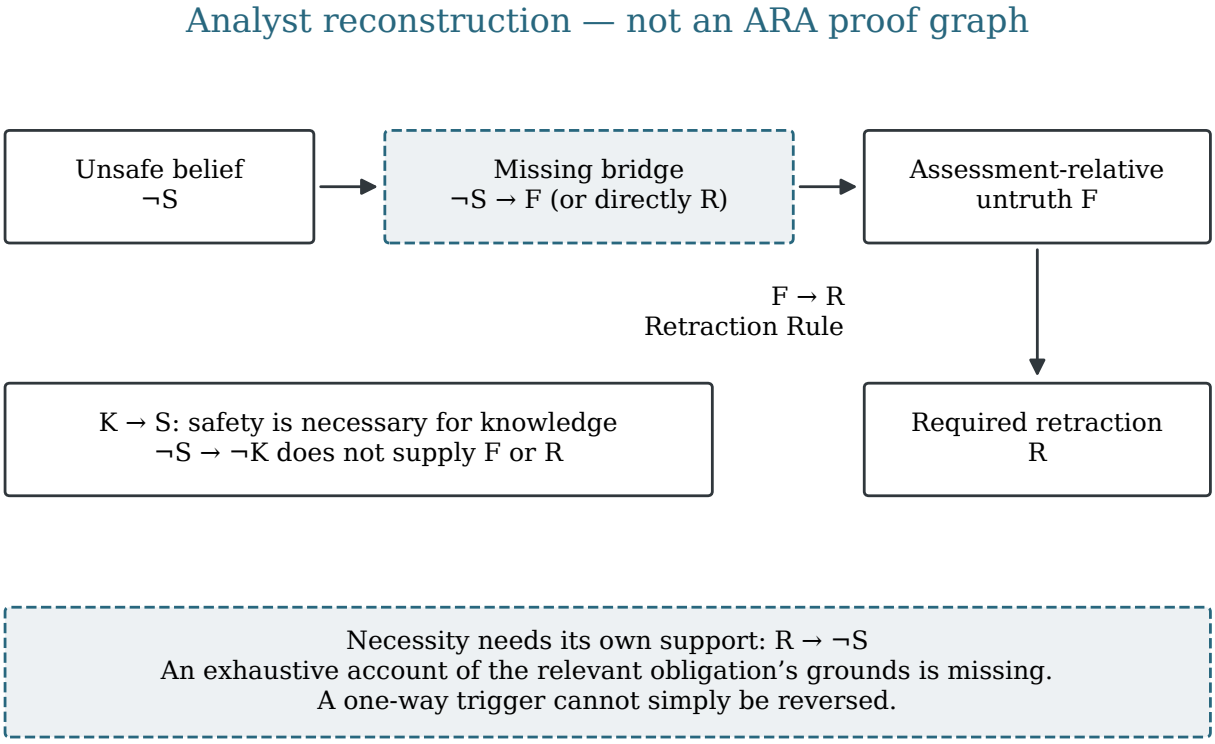


Figure 2. A schematic explanation of the missing links. The complete general Retraction Rule was absent from the original freeze and was retained in gap3; its inclusion did not establish the missing bridge. The diagram does not assume that the missing truth bridge is true or that it is the only possible strategy. All parameters must refer to the same belief, assertion and contexts.

MacFarlane's discussion of permissible retraction of something still thought true must not be used as a shortcut here. Permission and obligation differ. Likewise, a psychological account of why speakers retract does not establish the exhaustiveness of a constitutive norm. A future argument for necessity must engage those distinctions rather than infer a universal rule from a familiar example. [2, printed pp. 109, 111; 5, PDF pp. 10–11]

5.3 What the unsuccessful search warrants

The recorded searches support the claim that ARA did not license the supplied statements from this freeze within the tested procedure. The source inspection and the explicit rule structure support the missing-premise diagnosis. These are related but distinct grounds. Since C used whole-statement atoms and bounded search, its OPEN result alone cannot identify every possible missing premise or establish that the full literature entails neither polarity.

This also explains why calling T2 "novel" would overstate the evidence. A target can be absent from a bounded literature search, absent as a decomposed formula in T, and unsupported by a successful proof. Those absences do not combine into a positive contribution. At most they specify a research task whose completion would require an argument, faithful source interpretation and renewed comparison with the literature. [14]

6. Completed extensions: source coverage and the missing bridge

Gap2 reduced coverage to three source papers, retaining 57 commitments, and stopped before freezing a theory. Gap3 then retained all the identified L3 anchors, including the Beddor–Goldstein safety passages and MacFarlane's Retraction Rule. Its frozen theory contained 243 commitments from 23 of 26 papers, but no symbols were shared across source papers. Preserving the relevant passages therefore did not itself provide a shared formal connection between them. [14]

Gap3's packed continuation completed 46 polarity-search jobs over its 23 retained targets: all routes remained OPEN, with no accepted inference. The exact T2 probes were not admitted to this later target set. Its requests also omitted complete target bases, further limiting what can be inferred from the unsuccessful search. Consolidation subsequently accepted 22 batches comprising 181 commitments. A complete census of that accepted subset found zero symbols shared across source papers; the subset remained unfrozen. These outcomes leave the missing bridge unresolved rather than refuting the proposed connection. [14]

The separate T2 proposal paper explicitly argues for a bridge from later-assessed safety failure to an obligation to retract the same earlier assertion. It also addresses the additional exhaustiveness assumption needed for the converse. That paper develops the philosophical proposal; the connection has not been mechanically derived from the source commitments. [14]

7. What a defensible result would need

A positive T2 result would have to preserve a common speaker, belief, assertion token and pair of contexts across its premises. It would need a supported route from safety failure to the relevant obligation and an independently justified route back, with an explicit account of why the obligation has no safety-preserving exception within the stated domain. The formal representation would need to expose those directions instead of merely naming the entire thesis as an atom. Its source-linked proof and remaining critical questions would then need substantive review.

A weaker successful result could still matter. It might establish a suitably restricted sufficient condition, identify a counterexample to one direction, or show that a pragmatic rival handles the target cases without the proposed bridge. Such a result would need its own literature comparison; the dossier already anticipates several tempting reformulations. Labeling that outcome accurately would be more informative than treating every admitted target or polished essay as evidence for the exact biconditional.

The present outcome is narrower. ARA exposed a gap between selected source commitments and a proposed norm, and its focused test did not close it. The literature dossier shows both existing anticipations and a bounded failure to locate the exact T2 defense. This analysis explains the remaining obligations without fulfilling them by stipulation. The completed extensions retained the relevant source anchors but did not establish their shared formal connection or a licensed T2 derivation. The separate proposal paper makes the missing bridge explicit and offers an argument for it; assessing that argument remains distinct from closing an ARA route.

References

Bibliographic years follow the retained editions; online/issue-year differences are noted. “PDF p.” denotes a physical page in the verified local copy. The literature synopsis follows the page-checked dossier; the central safety, retraction and altered-factivity passages were also inspected directly for this paper. No long source quotations are reproduced.

1. Beddor, Bob, and Simon Goldstein (2021). “Mighty Knowledge.” *Journal of Philosophy* 118(5): 229–269.
2. MacFarlane, John (2014). *Assessment Sensitivity: Relative Truth and Its Applications*. Oxford University Press.
3. Peter Hawke (2024). *Stable acceptance for mighty knowledge*. https://doi.org/10.1007/s11098-024-02103-z
4. Peter Hawke (2024). *Modal Knowledge for Expressivists*. https://doi.org/10.1007/s10992-024-09759-2
5. Alexander Dinges (2021). *Knowledge and Asymmetric Loss*. https://doi.org/10.1007/s13164-021-00596-9
6. Bob Beddor; Andy Egan (2018). *Might do better: Flexible relativism and the QUD*. https://doi.org/10.3765/sp.11.7
7. Deniz Rudin (2024). *Asserting epistemic modals*. Online 2024; issue 2025. https://doi.org/10.1007/s10988-024-09423-7
8. Laura Caponetto (2024). *“Actually, Scratch That!”: A Tour into the Illocutionary Fabric of Retraction*. Inspected version: author preprint. https://doi.org/10.1007/978-3-031-66081-8_6
9. Lwenn Bussière-Caraes; Luca Incurvati; Giorgio Sbardolini; Julian J. Schlöder (2024). *Nevermind: On Retraction as a Speech Act*. https://doi.org/10.1007/978-3-031-66081-8_7
10. Derek Ball (2020). *Relativism, metasemantics, and the future*. https://doi.org/10.1080/0020174x.2020.1805710
11. Peter Hawke (2025). *Fragmentation and expression*. https://doi.org/10.1007/s11229-025-04915-w
12. Balder Edmund Ask Zaar (2025). *Does Safety Entail Context-Sensitivity?*. Online 2025; issue 2026. https://doi.org/10.1007/s10670-025-00960-6
13. Holliday, Wesley H., and Matthew Mandelkern (2024). “The Orthologic of Epistemic Modals.” *Journal of Philosophical Logic* 53. Author manuscript: https://arxiv.org/abs/2203.02872.
14. *ARA run records* (2026). Literature-comparison dossier, source commitments, argument-search results, accepted-subset vocabulary census, and the T2 proposal manuscript. Records examined through 1 November 2026.

Status and method

AI-generated analysis of ARA output; not an ARA-derived route.